

Convolution Theorem

$$\mathcal{L}\{f(t) * g(t)\} = F(s) G(s)$$

$$\text{where: } f(t) * g(t) = \int_0^t f(\theta) g(t-\theta) d\theta$$

Proof:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$G(s) = \mathcal{L}\{g(t)\} = \int_0^{\infty} e^{-st} g(t) dt$$

The variable t in the integrals above is a dummy variable, we can replace it by any variable.

$$\begin{cases} F(s) = \int_0^{\infty} e^{-s\theta} f(\theta) d\theta \\ G(s) = \int_0^{\infty} e^{-sx} g(x) dx \end{cases}$$

$$\begin{aligned}
 F(s)G(s) &= \int_0^{\infty} e^{-s\theta} f(\theta) d\theta \int_0^{\infty} e^{-sx} g(x) dx \\
 &= \int_0^{\infty} \int_0^{\infty} f(\theta) g(x) e^{-s(x+\theta)} dx d\theta
 \end{aligned}$$

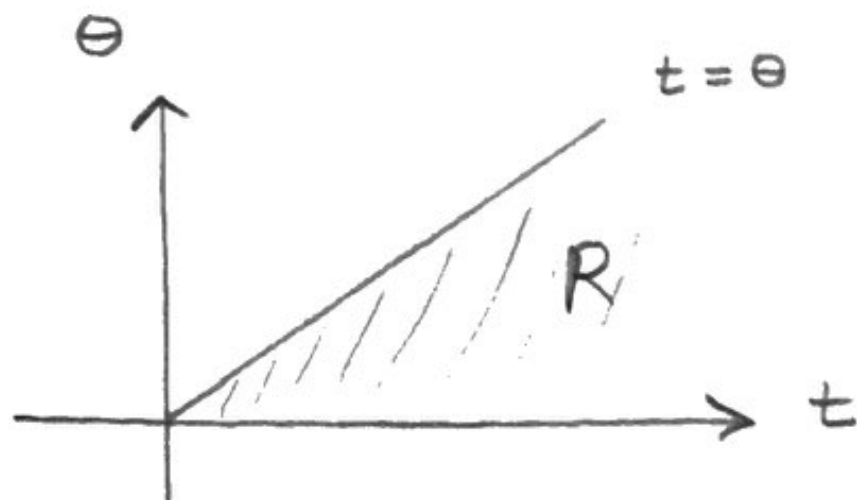
For a fixed θ , let $t = x + \theta \rightarrow dt = dx$

$$\begin{cases} x = \infty \rightarrow t = \infty \\ x = 0 \rightarrow t = \theta \end{cases}$$

$$F(s)G(s) = \int_0^{\infty} \int_{\theta}^{\infty} f(\theta) g(t-\theta) e^{-st} dt d\theta$$

The region of integration of the double integral is

$$R : \begin{cases} 0 \leq \theta < \infty \\ \theta \leq t < \infty \end{cases}$$



R can also be described as :-

$$R : \begin{cases} 0 \leq t < \infty \\ 0 \leq \theta \leq t \end{cases}$$

$$\Rightarrow F(s)G(s) = \int_0^{\infty} \int_0^t f(\theta) g(t-\theta) e^{-st} d\theta dt$$

Swap the order

$$\Rightarrow F(s)G(s) = \int_0^{\infty} e^{-st} \left(\int_0^t f(\theta) g(t-\theta) d\theta \right) dt$$

$f(t) * g(t)$

$$\therefore F(s)G(s) = \mathcal{L}\{f(t) * g(t)\}$$